

Composing transformations



1

a Reflection and translation

b Rotation and translation

2

a Reflection and translation

b Rotation and translation

3 Shape A

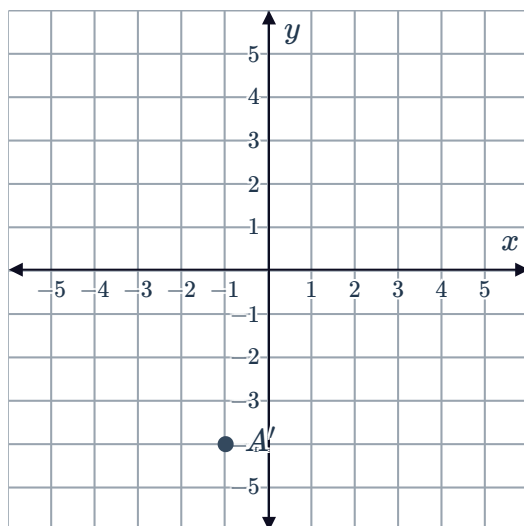
4 Triangle W

5 Square A

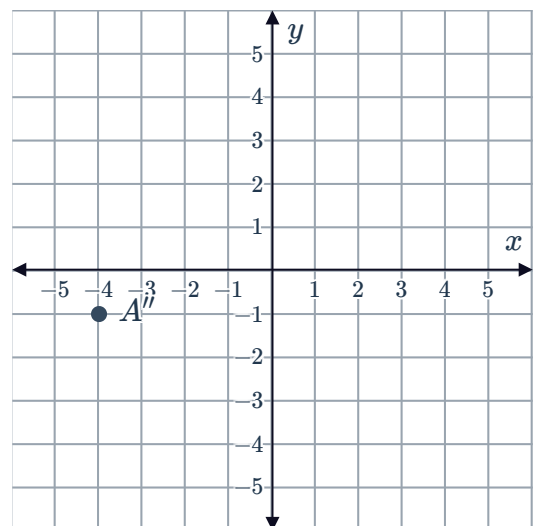
6 $CBAD$

7

a

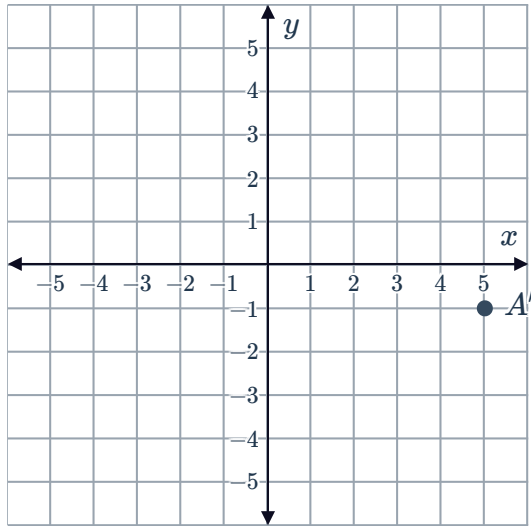


b

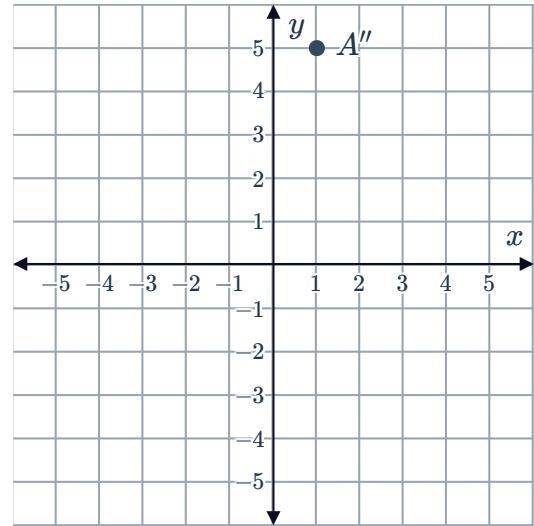


8

a

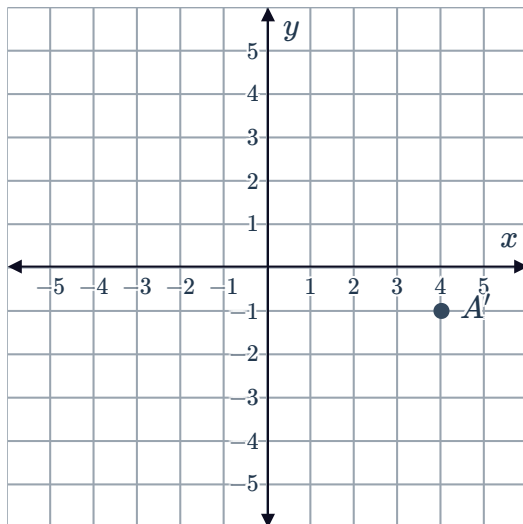


b

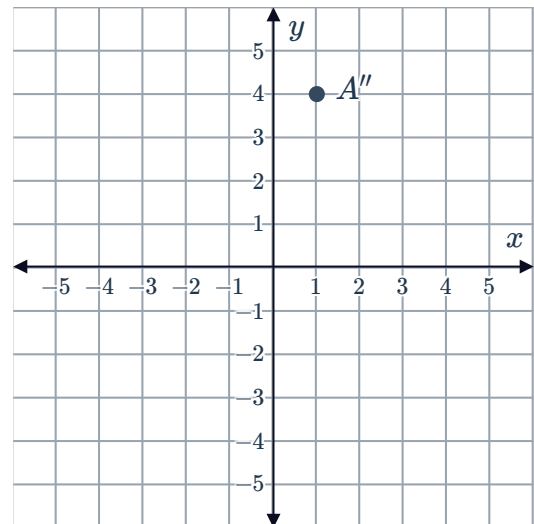


9

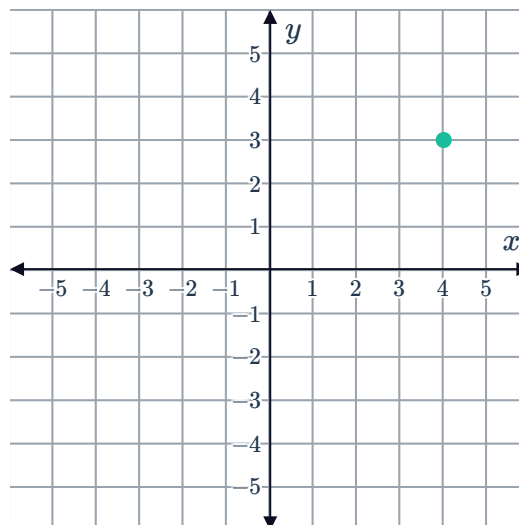
a



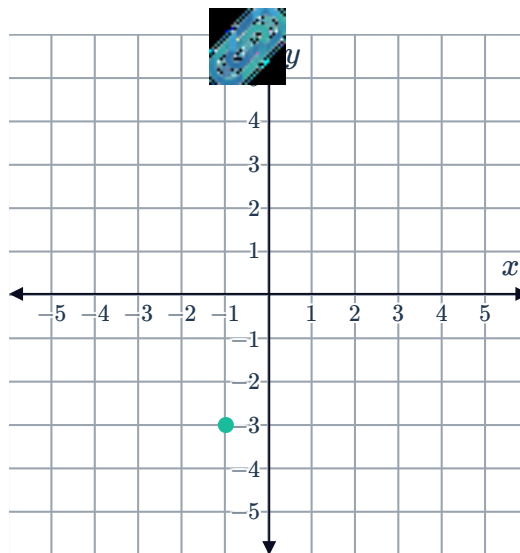
b



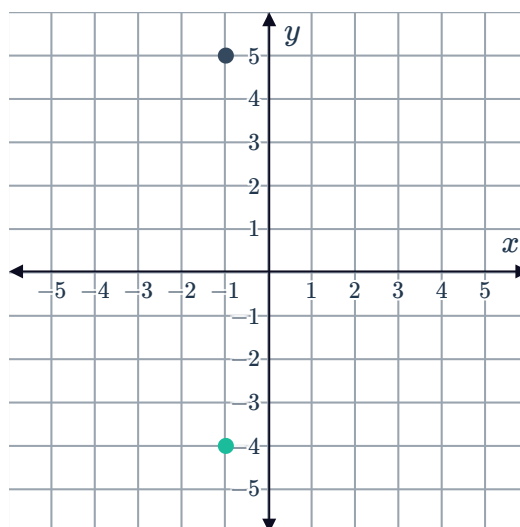
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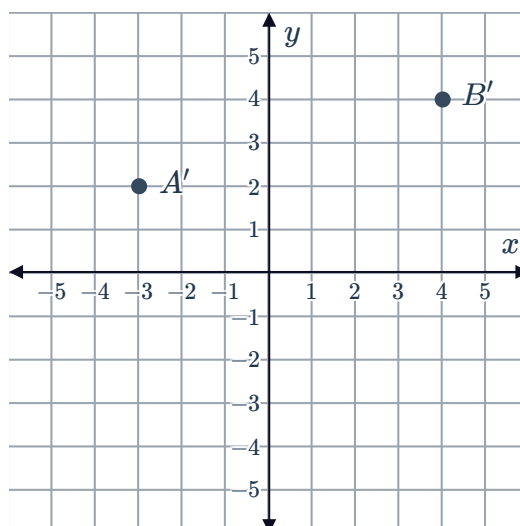
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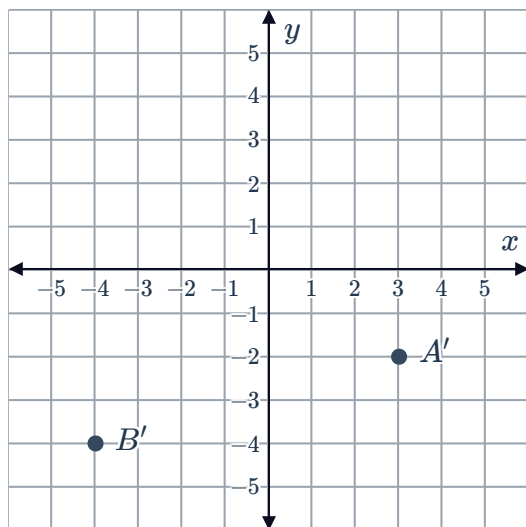
12



13

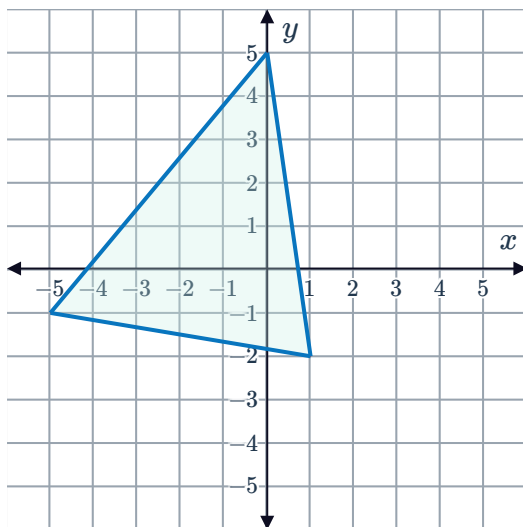


14
a

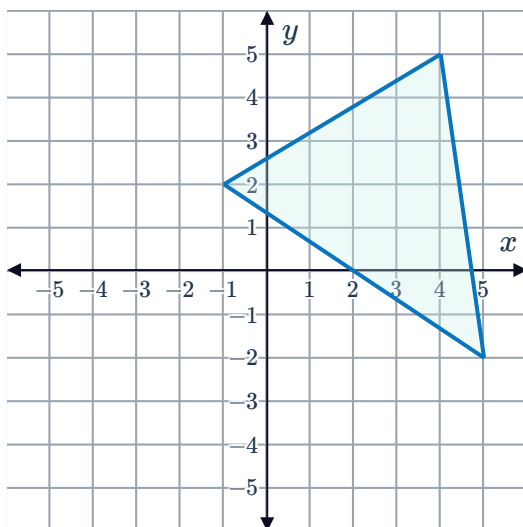


b A reflection across the x -axis.

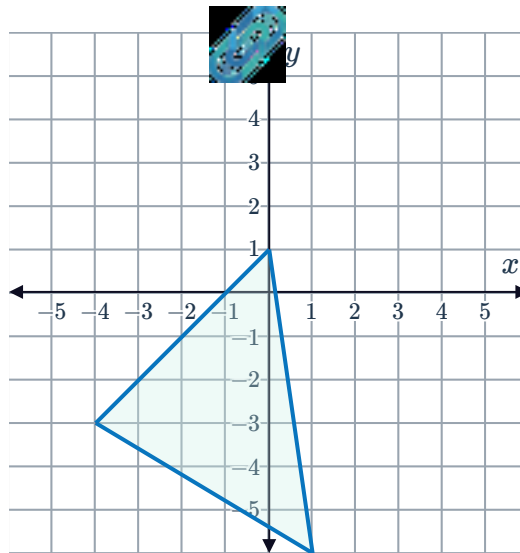
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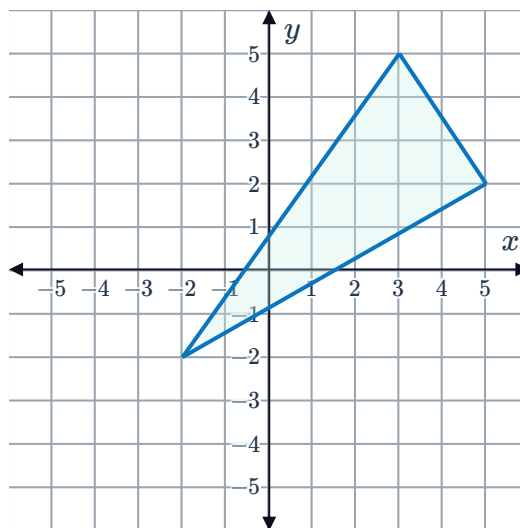
16



17

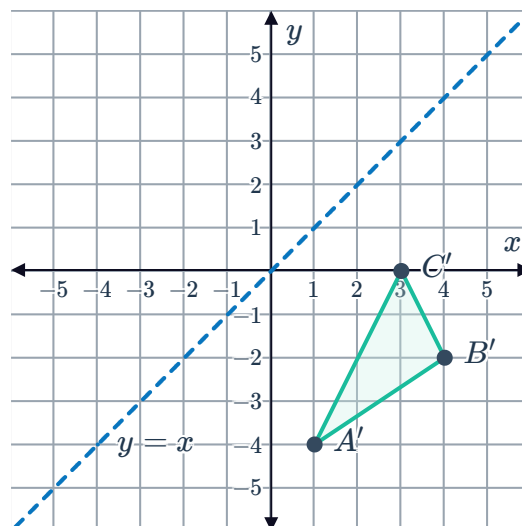


18



19

a



b A 90° rotation anticlockwise about the origin.

20

a $B(4, -1)$ b $y = 1$



21 A reflection across the y -axis, and a translation 10 units left and 11 units up.

22 A translation 11 units right and 3 units up, and a reflection across the x -axis.

23 A reflection across the x -axis, and a translation 10 units left and 8 units down.

24 A translation 9 units right and 8 units up, and a reflection across the x -axis.

25

a $B(6, 10)$

b 16 units

26 $(9, 10)$

27 $(1, 2)$

28

a $(-5, 6)$

b $(4, -7)$

29 $(1, 3)$

30 $(11, 7)$

31

a $A'(-6, -1)$

b A reflection across the y -axis.

32

a $A'(4, -4), B'(-8, 4)$

b A reflection across the y -axis.

33

a $A'(-1, -2), B'(-7, 3)$

b A reflection across the x -axis.

34 $A'(-8, 2), B'(-5, 9), C'(-14, 6)$

35 $A'(5, 4), B'(-1, -1), C'(2, -10)$

36 $A'(2, -8), B'(-4, 5), C'(-3, 6)$

37 $A'(9, -5), B'(-10, 1), C'(-10, -3)$



39 $A'(-1, -9), B'(-7, -2), C'(-6, 4)$

40 $(-x, -y)$

41 Reflecting the point across both the x and y -axes.

42 $(a, -b)$

43

a $B(0, 9)$

b 9 unit translation horizontally to the right, and 9 unit translation vertically upwards.

44

a $C(5, -6)$

b Rotation 180° about the origin.